Microsoft Elevate

CAPSTONE PROJECT

AI-Based Housing Price Prediction Using Machine Learning

PRESENTED BY

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PROBLEM STATEMENT:

Accurate housing price prediction is essential for buyers, sellers, and real estate investors. Traditional methods rely on manual estimation and market intuition, which can be inaccurate and inconsistent.
With the availability of structured datasets, machine learning can be leveraged to build predictive models that estimate housing prices based on various socio-economic and geographical features.

PROPOSED SOLUTION:

- The proposed system utilizes supervised machine learning to predict housing prices using the California Housing dataset.
- The workflow includes data preprocessing, feature selection, model training, and evaluation.
- A Random Forest Regression model is employed to learn patterns between housing attributes and price values.
- The trained model generates accurate price predictions and is evaluated using metrics such as Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) to ensure reliability and performance.

SYSTEM APPROACH:

Dataset → Data Preprocessing → Model Training → Evaluation → Prediction Output

The system processes raw housing data, performs preprocessing, trains machine learning models, evaluates performance, and generates predictions.

ALGORITHM & DEPLOYMENT:

- **Algorithm Selection**

A **Random Forest Regression** algorithm is chosen for predicting housing prices. It is selected due to its ability to model complex, non-linear relationships and provide high accuracy on structured tabular datasets.

- **Data Input**

The model uses features from the California Housing dataset, including:

➔ Median income

➔ House age

➔ Average rooms and bedrooms

➔ Population and households

➔ Geographic location (latitude and longitude)

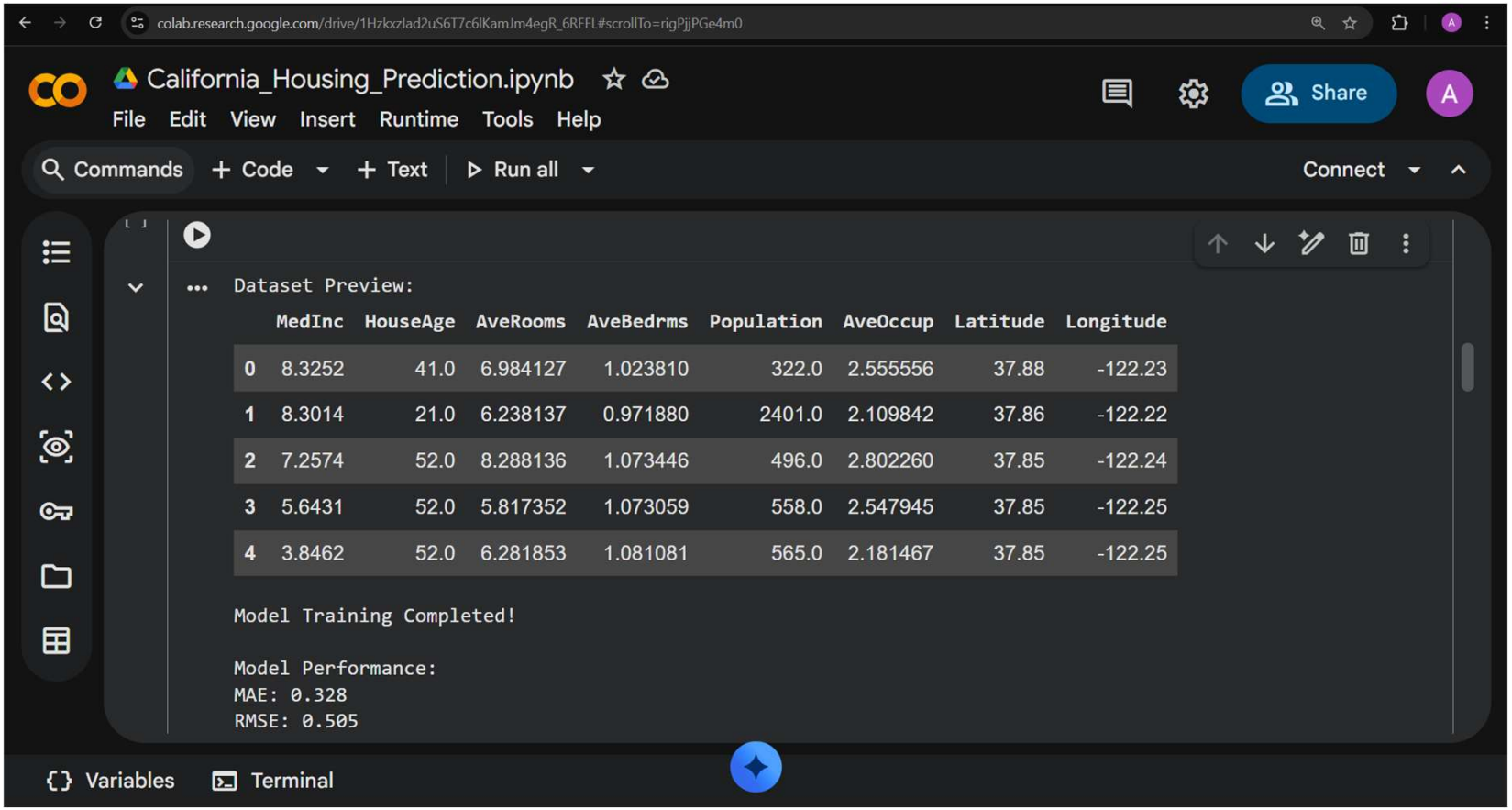
- **Training Process**

The dataset is split into training and testing sets. The Random Forest model is trained on historical housing data, and performance is validated using standard evaluation metrics to ensure reliability.

- **Prediction Process**

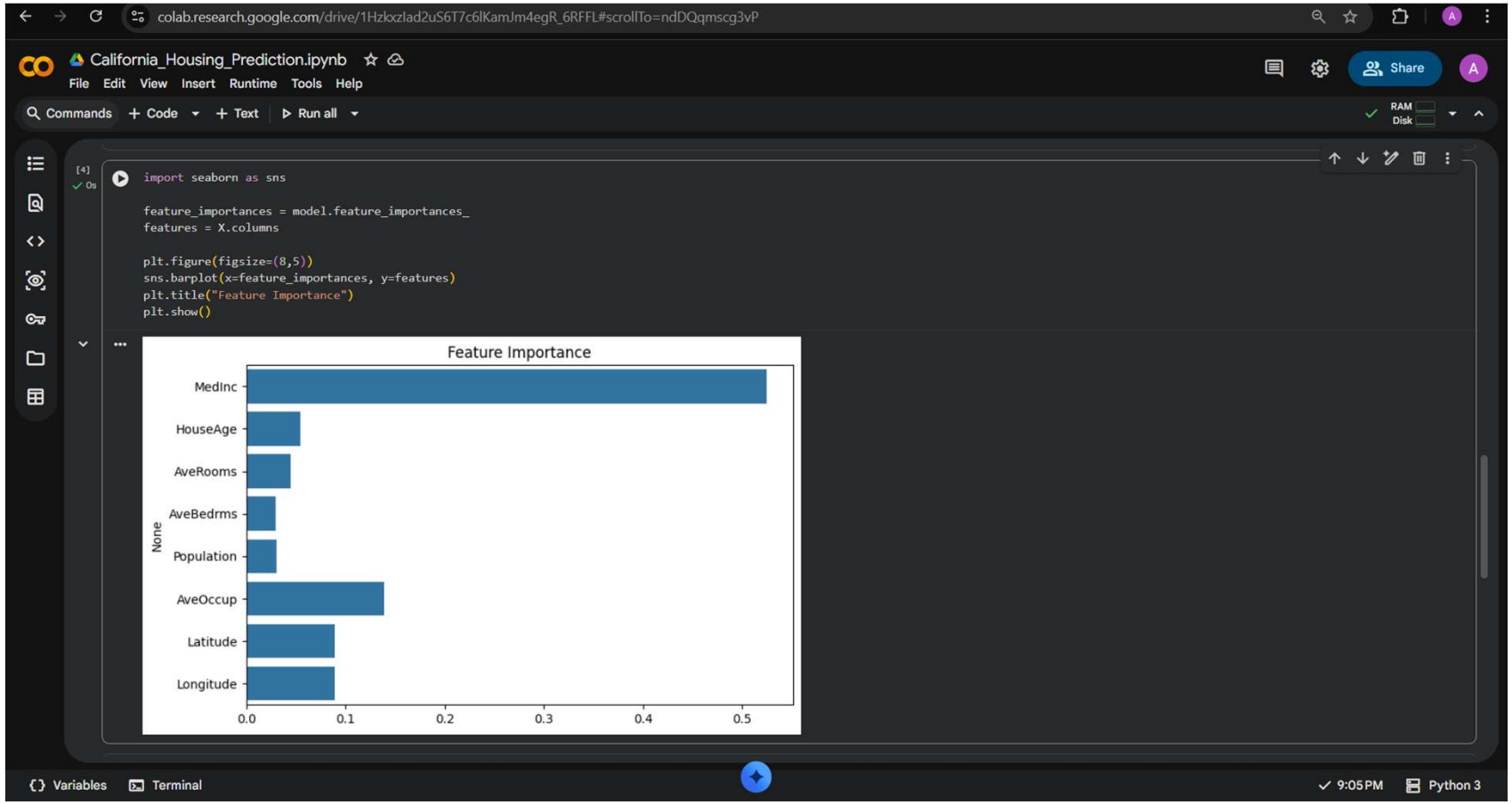
After training, the model predicts housing prices for unseen data by analyzing feature patterns. The predictions help estimate property values accurately based on socio-economic and location-based factors.

Dataset Overview:



	MedInc	HouseAge	AveRooms	AveBedrms	Population	AveOccup	Latitude	Longitude
0	8.3252	41.0	6.984127	1.023810	322.0	2.555556	37.88	-122.23
1	8.3014	21.0	6.238137	0.971880	2401.0	2.109842	37.86	-122.22
2	7.2574	52.0	8.288136	1.073446	496.0	2.802260	37.85	-122.24
3	5.6431	52.0	5.817352	1.073059	558.0	2.547945	37.85	-122.25
4	3.8462	52.0	6.281853	1.081081	565.0	2.181467	37.85	-122.25

Feature Importance Analysis:



Model Training & Evaluation:

- Model trained using Random Forest Regressor
- Data split into training and testing sets
- Achieved low MAE and RMSE values
- Indicates strong prediction accuracy

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Model Training Completed!

Model Performance:
MAE: 0.328
RMSE: 0.505
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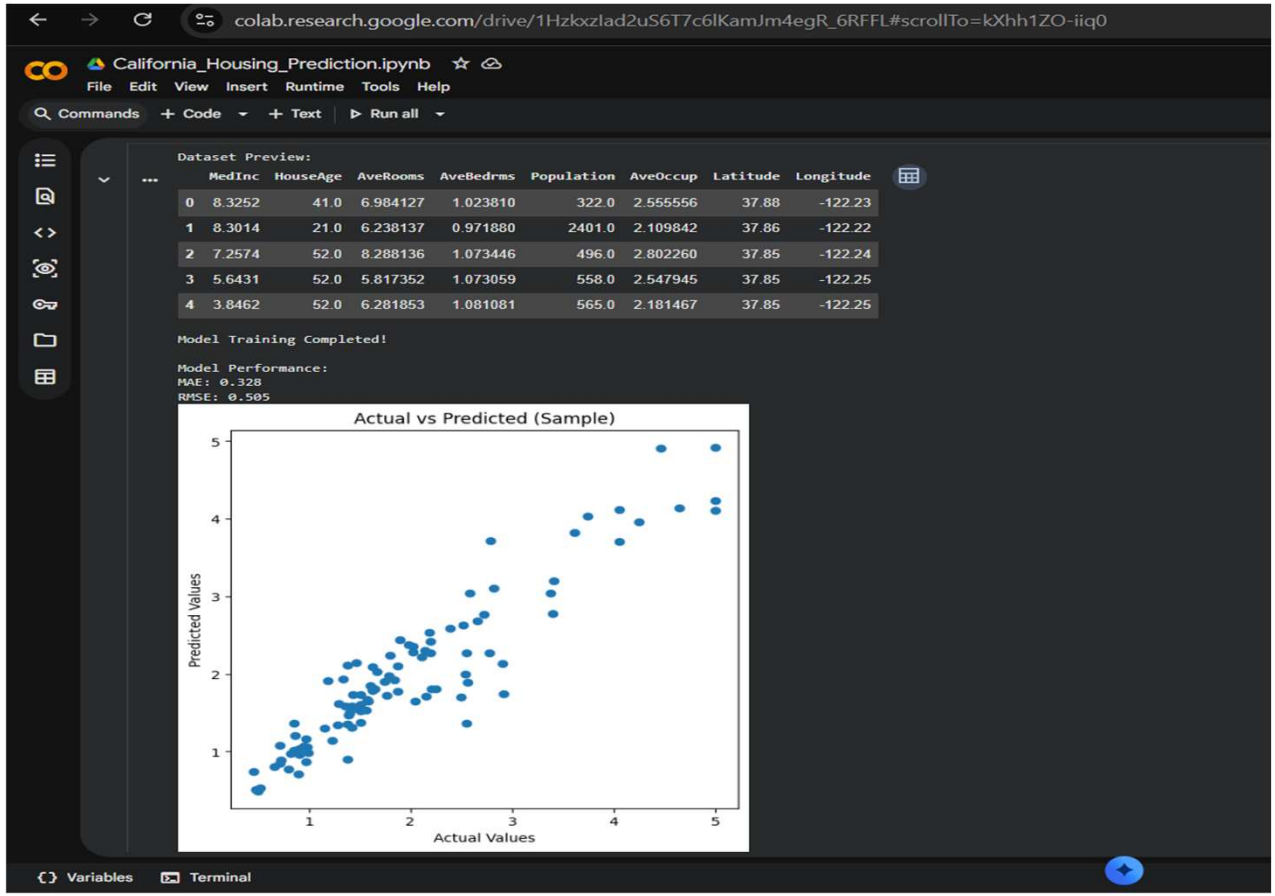
Prediction Visualization:



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RESULT:

The model successfully learned relationships between socio-economic features and housing prices.
Ensemble learning techniques provided better accuracy than linear models.
The system demonstrates the effectiveness of machine learning in real-world prediction problems.

CONCLUSION:

A machine learning-based housing price prediction system was successfully developed and evaluated.
Random Forest achieved strong performance with low prediction error.
The project demonstrates the potential of AI in real estate analytics and data-driven decision making.

FUTURE SCOPE:

- Integration with real-time housing data
- Deployment as a web application
- Use of deep learning models for improved accuracy
- Expansion to multi-city datasets
- Integration with cloud platforms like Microsoft Azure

REFERENCES:

- California Housing Dataset
- Scikit-learn Documentation
- Machine Learning Research Articles

GitHub Link: GitHub.com/adarshkumar-core/California-Housing-AI-Project

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Thank You